Research Idea: GS1354-645

ALMA Thesis Planner (automated pipeline)

Title: How steady is the millimeter jet? A homogeneous uv-plane flux and variability-limit measurement of the ALMA black-hole X-ray-binary ToO program (2013.1.00222.T), with self-calibration treated as a measurement problem

What the thesis delivers. The headline is a *measurement*, not a verdict: homogeneous visibility-plane millimeter flux densities for every execution block this ToO program fired, plus a quantitative fractional-rms variability curve (or limit) as a function of timescale, with a systematics floor that is characterized rather than asserted. A separate theory chapter derives what the competing jet pictures actually predict at 230 GHz and at minute-to-hour timescales, so the comparison at the end rests on a calculation the student did, not on numbers borrowed from intuition.

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Step 0 — audit, one to two weeks, before any science framing is fixed. Query the ALMA Science Archive for everything under 2013.1.00222.T and tabulate per execution block: source triggered, date, receiver band, on-source time, array configuration, spectral setup, phase/bandpass/flux calibrators, presence or absence of a check source, and a quick continuum detection and flux density from the delivered calibrated data. **In parallel, and equally binding: check the published literature for the source(s) actually triggered** — are there published radio flux densities near that date, and a published X-ray state classification for that outburst? Record what exists and what does not. The realistic expectation is one to three blocks on a single unresolved source, probably Band 6, likely tens of mJy; the thesis is written for that minimal case. A second well-separated band, multiple epochs, or a check source are upside. This audit table is the first deliverable and it fixes the achievable scope of everything after it.

Chapter A — the measurement pipeline. 1. *Restore and inspect.* Re-run the delivered CASA calibration; examine gain, bandpass and $T_{\{}rm sys}$ solutions and the atmospheric phase RMS; record observing conditions per block. 2. *Visibility-plane point-source photometry.* Fit the target as an unresolved point source directly in the uv plane, per time bin and per sideband, rather than by image-plane fitting. This is what makes low-SNR fine time-binning possible at all. 3. *Bin length set by measured noise*, per block, so each bin reaches SNR $\approx$ 5–10; the resulting time resolution and detectable fractional-rms amplitude are stated *before* any variability result is quoted.

Chapter B — self-calibration as a first-class methodological question. For a tens-of-mJy point source in Band 6, phase-only self-calibration is the standard and most effective defence against atmospheric decorrelation, and it is also the step most likely to eat the signal being measured. The thesis treats this explicitly rather than choosing a side:

- Run the full photometry **twice**: once through standard phase-referenced calibration only, once with phase-only selfcal (amplitude selfcal is never used, since it directly absorbs source flux variation). Compare fluxes, per-bin scatter, and recovered rms between the two paths.

- **Injection–recovery is the core test.** Inject synthetic flux modulations into the calibrated visibilities — sinusoids and shot-like flares spanning timescales from below to well above the selfcal solution interval, at amplitudes from a few percent to tens of percent — and push them through each pipeline. The output is a measured **transfer function $T(\{}tau)$**: the fraction of true fractional rms at timescale $\{}tau$ that survives to the final light curve. This converts selfcal from a worry into a calibrated, quantified filter, lets solution intervals be chosen on evidence, and allows every reported rms and every upper limit to be corrected for, or explicitly bounded by, the pipeline's own suppression. It also settles empirically whether selfcal's decorrelation gain outweighs its signal loss at each timescale for this data.
- Independent systematics bounds, kept as bounds: a check source through the identical pipeline where one exists (the only genuine control); otherwise gains solved on the phase calibrator and transferred to the bandpass/flux calibrator treated as an unknown, plus odd-to-even phase-cal scan transfer. Both are stated as **loose upper bounds, not corrections** — odd/even doubles the interpolation interval and so overestimates decorrelation, and calibrators are 10–100× brighter and at different elevation and sky position. Where no check source exists, every variability statement is framed as a limit.

Chapter C — what the pictures actually predict (the derivation, ~4–6 weeks). Rather than asserting rms values, the student computes them for this source's plausible parameters:

- *Compact conical jet (Blandford–Königl-type).* Locate the $\{}tau_\{}nu = 1$ photosphere, using $z_\{}nu \{}propto \{}nu^{-1}$ anchored to the source's published radio flux and distance. Notably, the honest answer is that a Band 6 photosphere is of order 10^{10}–10^{11} cm — light-*seconds*, not light-minutes — so smearing does **not** automatically suppress minute-scale variability. The derivation therefore yields a low-pass filter with a corner at seconds-to-tens-of-seconds, meaning mm emission should track fluctuations in jet-base injection with only modest damping on the timescales this data can probe.
- *Internal shocks / discrete ejecta.* Compute the collision radius and flare rise time for shells launched with a Lorentz-factor spread and a launching cadence set by inner-accretion-flow variability, and the resulting light-curve morphology.
- The output is a predicted **fractional-rms-versus-timescale curve for each picture**, with its parameter sensitivity shown, plus their distinguishing signature: not a single rms number, but whether the variability is smooth red-noise wander (filtered injection) or discrete flare structure, and at which timescale power appears. This is what the measurement is compared against.

In-band spectral index: a stated limitation, not a result. A Band 6 continuum setup spans ~224–240 GHz, $\{}ln(\{}nu_2/\{}nu_1) \{}approx 0.07$, so $\{}sigma_\{}alpha \{}approx 1.4\{},(\{}sigma_F/F)/0.07$ — recovering $\{}sigma_\{}alpha \{}approx 0.3$ demands ~1.5% *relative* sideband accuracy, which thermal noise permits but sideband bandpass and gain-transfer systematics do not. The student writes this budget out explicitly, reports the sideband flux ratio with the systematics term dominant, and states that the whole flat-to-inverted range of interest sits inside the error bar. A band-to-band index is measured **only if** the audit reveals a genuine second, well-separated band, where the lever arm makes absolute flux-scale precision the limiting term instead.

Astronomical interpretation, scoped to what exists. Whatever the audit found in the literature — published radio flux densities near the trigger date and an X-ray state classification — is used by ordinary citation to place the mm flux (or upper limit) relative to the extrapolated flat radio spectrum, and hence to bound where the optically thick–thin break can lie. If Step 0 shows that record is thin for the triggered source, the interpretation is scaled back accordingly and said so. The separately-awarded quasi-simultaneous radio, optical-IR, Swift/INTEGRAL/Suzaku/NuSTAR and Fermi observations are **not** part of this dataset; no conclusion rests on reduced products from them, and full broadband SED modeling and jet-power fitting are named as the follow-up those data would enable.

Deliverables. (a) The program audit table plus a literature-availability audit for the triggered source. (b) A documented, reusable uv-plane photometry pipeline with both calibration paths. (c) Homogeneous flux densities per block and per sideband, with statistical error, absolute flux-scale error quoted separately, and the bounded systematics floor. (d) The measured selfcal transfer function $T(\{}tau)$ — a transferable methods result usable by anyone doing ALMA photometry of faint variable point sources. (e) Light curves with pre-stated, transfer-function-corrected detection thresholds, and fractional-rms limits or detections across the accessible timescales. (f) The derived rms-versus-timescale predictions for both jet pictures, and a comparison stating plainly which timescales the data can and cannot discriminate on.

If the source is faint or undetected. The thesis becomes a deep, carefully bounded mm upper limit placed on the published radio-to-IR spectrum of that outburst — constraining how far the flat compact-jet spectrum extends and where the break must lie — with the injection–recovery and selfcal characterization, run on the calibrators, standing as the methods contribution. Neither branch leaves the thesis empty.

Why it is thesis-sized, and why it is not thin. A bounded handful of short blocks on one unresolved point source: no deconvolution, no confusion, no external catalog dependency, no new observations. The volume comes from three real pieces of work — a careful low-SNR photometry pipeline, a quantitative selfcal injection–recovery study, and a first-principles derivation of what mm variability each jet model predicts — which together fill a semester-to-a-year without any of them being open-ended.